



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2004

YEAR 12

HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK # 3

Mathematics

General Instructions

- Working time – 2 hours.
- Reading Time – 5 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work.
- Hand in your answer booklets in **3** sections.
- Start each **NEW** section in a separate answer booklet.

Total Marks - 90

- Attempt Sections A – C
- All sections are of *equal* value.

Examiner: *P. Parker*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate.

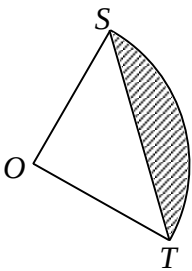
Total marks – 90

Attempt Questions 1 – 6

All questions are of EQUAL value

Answer each section in a SEPARATE writing booklet. Extra writing booklets are available.

SECTION A (Use a SEPARATE writing booklet)

Question 1 (15 marks)		Marks
(a)	Write down the value of $\cos 1.5$, correct to 3 decimal places.	1
(b)	Find $\int (3x^2 + 1) dx$	1
(c)	Sketch, showing intercepts with the coordinate axes, $y = 2 \cos x \text{ for } 0 \leq x \leq 2\pi$	2
(d)	If $f(x) = 3 \cos 2x$, find the exact values of	
(i)	$f\left(\frac{\pi}{6}\right)$	2
(ii)	$f'\left(\frac{\pi}{6}\right)$	2
(iii)	$f''\left(\frac{\pi}{6}\right)$	2
(iv)	$\int_0^{\frac{\pi}{6}} f(x) dx$	2
(e)	The diagram below shows a sector of a circle with centre O and radius 4 cm and $\angle TOS = \frac{\pi}{6}$.	
		
(i)	Find the length of the arc ST .	1
(ii)	Find the area of sector OST , leaving your answer in terms of π .	1
(iii)	Hence, or otherwise, calculate the shaded area correct to 2 decimal places.	1

Question 2 (15 marks)

Marks

- (a) Find the equation of the normal to the curve $y = 4 \ln x$ at the point where $x = 1$. 2
- (b) Simplify $e^{2 \ln 4}$. 1
- (c) The displacement (x m) of an object (at time t secs) is given by the equation
$$x = t^3 - 15t^2 + 48t - 25$$

(i) Find expressions for the object's velocity and acceleration. 2
(ii) Find its initial displacement, velocity and acceleration. 3
(iii) When is the object at rest? 2
- (d) Using the Table of Integrals provided, find the exact value of
$$\int_0^{\frac{\pi}{8}} \sec 2x \tan 2x \, dx$$
 2
- (e) Which of the following areas is larger? Justify your answer 3

Area 1 is bounded by $y = 1.5 + \sin x$, x axis,
 $x = 0$ and $x = \pi$.

Area 2 is bounded by $y = 1.5 + x^2$, x axis,
 $x = 0$ and $x = \pi$.

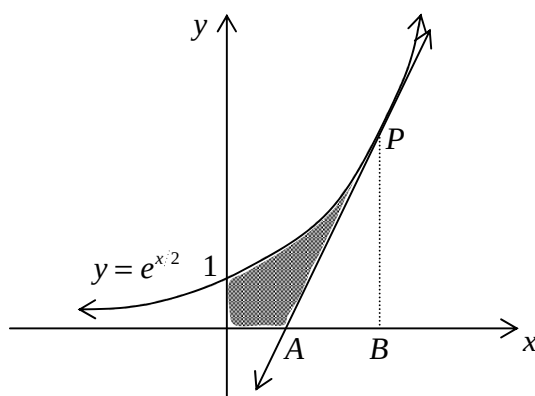
SECTION B (Use a SEPARATE writing booklet)

Question 3 (15 marks)

Marks

- (a) Find $\frac{dy}{dx}$ given
- | | | |
|-------|----------------------------|---|
| (i) | $y = e^{-x} \tan x$ | 2 |
| (ii) | $y = \ln(x^4) + (\ln x)^4$ | 2 |
| (iii) | $y = \frac{x^2}{\ln x}$ | 2 |

- (b) The diagram below shows the graph of $y = e^{x/2}$, together with a tangent line AP .
 B is the point $(3, 0)$ and P is the point $(3, e^{3/2})$



- | | | |
|-------|---|---|
| (i) | Find the equation of the tangent AP . | 2 |
| (ii) | Show that A has coordinates $(1, 0)$. | 1 |
| (iii) | Hence, or otherwise, find the shaded area in terms of e . | 2 |
-
- | | | |
|-----|---|---|
| (c) | Find $\int \left(\frac{e}{x} + \frac{x}{e} \right) dx$ | 2 |
| (d) | Differentiate $\ln(e^2 x^2)$ | 2 |

Question 4 (15 marks)

Marks

- (a) The height of the tide at Sunset Beach can be modelled by the equation

$$H = 4 + 2 \cos \frac{\pi t}{6}, \quad t > 0$$

where H is the height of the tide in metres after time t hours.

- (i) Find the rate of change of height of the tide after 10 hours. 2
- (ii) State whether the tide was an incoming or outgoing tide after 10 hours (give reasons). 1

- (b) *“Whilst the price of houses continues to increase, recent moves by the State Government aimed at reducing the price of houses seem to be taking effect.”*

Given that P is the price of houses in Australian dollars, what does the above statement say about:

- (i) $\frac{dP}{dt}$? 1
- (ii) $\frac{d^2P}{dt^2}$? 1

- (c) The rate of decay of a radioactive element is proportional to the mass M of the radioactive element present. The process is described by the differential equation

$$\frac{dM}{dt} = -kM, \quad \text{where } k > 0$$

- (i) Show that $M = M_0 e^{-kt}$ satisfies this differential equation where M_0 is a constant. 1
- (ii) Fifty percent of the original mass of the radioactive element Strontium 90 remains after 28 years. Show that $k \cong 0.02476$. 2
- (iii) How long will it take for Strontium 90 to decay to 1% of its original mass? 2

- (d) The function $f(x) = x^{-e}e^x$, where $x > 0$

- (i) Show that $f'(x) = e^x x^{-(e+1)}(x - e)$. 3
- (ii) Classify the turning point at $x = e$. Justify your answer. 2

SECTION C (Use a SEPARATE writing booklet)

Question 5 (15 marks)

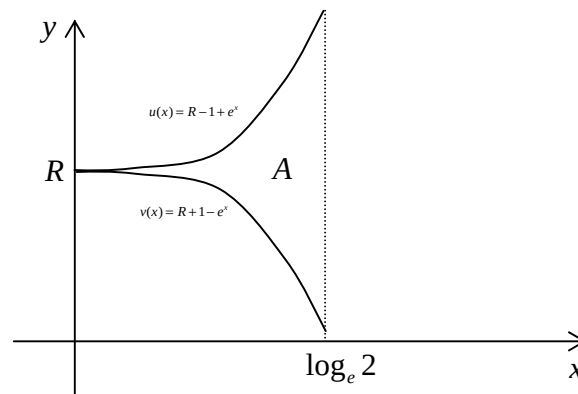
Marks

- (a) The velocity of a particle traveling in a straight line is given by

$$v = e^{3t} - 4e^t \text{ m/s}$$

where t is in seconds and $t \geq 0$.

- | | | |
|-------|--|---|
| (i) | Find the initial velocity and acceleration. | 2 |
| (ii) | Find the value of t when the particle is stationary. Give your answer in the form $\ln k$, where k is a constant. | 2 |
| (iii) | If initially the particle is at $x = 0$, find an expression for the position x in terms of t . | 2 |
| (iv) | Show that at $t = \ln 4$ seconds the particle is 9 metres to the right of the origin. | 1 |
| (v) | Find the distance travelled by the particle in the first $\ln 4$ seconds | 2 |
- (b) In the diagram below $u(x) = R - 1 + e^x$ and $v(x) = R + 1 - e^x$



[Note: Diagram is not drawn to scale, $R > 1$]

- | | | |
|------|---|---|
| (i) | Show that A , the area between the curves $u(x)$ and $v(x)$ where $0 \leq x \leq \log_e 2$ is given by | 3 |
| | $A = 2(1 - \log_e 2)$ | |
| (ii) | A solid is formed by rotating the area between the curves $u(x)$ and $v(x)$ where $0 \leq x \leq \log_e 2$, about the x axis. Show that the volume of the solid generated is $2\pi RA$ where A is the area between the curves. | 3 |

Question 6 (15 marks)

Marks

- (a) Evaluate

3

$$\int_{-1}^1 \left(2x - \frac{1}{x-2} \right) dx$$

Express your answer in the form $\ln k$, where k is a real constant.

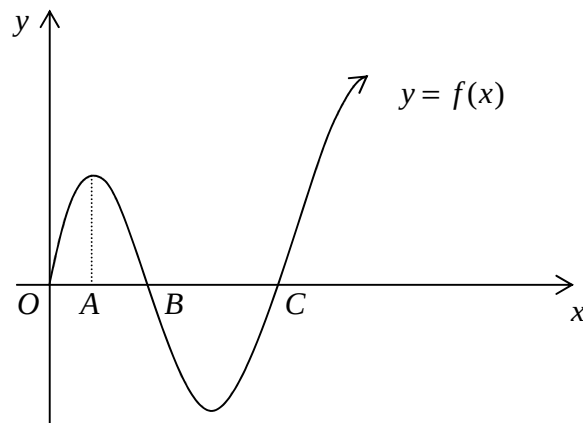
- (b)

- (i) If $f(x) = e^{3x} \sin 2x$ show that $f'(x) = 0$ when $\tan 2x = -\frac{2}{3}$

2

The diagram below shows the graph of $f(x) = e^{3x} \sin 2x$ for $x \geq 0$.

This graph intersects the x axis at the points B and C . The point A is a local maximum.



- (ii) (α) Show that the x coordinates of B and C are $\frac{\pi}{2}$ and π respectively.

2

- (β) Find the x coordinate of A , correct to 2 decimal places.

2

- (iii) (α) If $g(x) = \frac{1}{13} (3e^{3x} \sin 2x - 2e^{3x} \cos 2x)$ show that

3

$$g'(x) = e^{3x} \sin 2x$$

- (β) Hence or otherwise, find the area between the graph of $f(x) = e^{3x} \sin 2x$ and the x axis from O to B .

3

End of paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$



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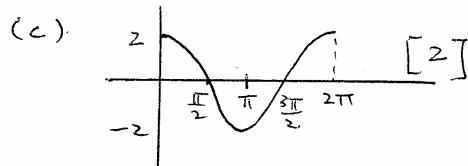
Sample Solutions

SECTION	MARKER
A	Mr Choy
B	Mr Gainford
C	Mr Fuller

Question (1).

(a) $6 \rightarrow 1.5^6 = 0.071$ [1]

(b) $\int (3x^2 + 1) dx$
 $= x^3 + x + C$ [1]



$y = 2 \cos x, 0 \leq x \leq 2\pi$

(d) $f(x) = 3 \cos 2x,$

(i) $f(\frac{\pi}{6}) = 3 \cos \frac{\pi}{3}$
 $= \frac{3}{2}$ [2]

(ii) $f'(\frac{\pi}{6}) = -6 \sin \frac{\pi}{3}$
 $= -3\sqrt{3}$ [2]

(iii) $f''(\frac{\pi}{6}) = -12 \cos \frac{\pi}{3}$
 $= -6$ [2]

(iv) $\int_0^{\pi/6} f(x) dx = 3 \int_0^{\pi/6} \cos 2x dx$
 $= \left[\frac{3 \sin 2x}{2} \right]_0^{\pi/6} = \frac{3\sqrt{3}}{4}$ [2]

(e) (i) $l = 4 \times \frac{\pi}{6}$
 $= \frac{2\pi}{3}$ [1]

(ii) $A = \frac{1}{2} \times 16 \times \frac{\pi}{6}$
 $= \frac{4\pi}{3}$ [1]

(iii) Area of minor segment
 $= \frac{4\pi}{3} - 4 = 4(\frac{\pi}{3} - 1)$
 $= 0.19$ (2 d.p.) [1]

Question (2).

(a) $y = 4 \ln x.$

$\frac{dy}{dx} = \frac{4}{x}, \frac{dy}{dx} \Big|_{x=1} = 4$
 $x=1, y=0 \quad (1,0) \quad m=4$

$y - 0 = 4(x - 1)$
 $4y = -x + 1$ [2]

$x + 4y - 1 = 0$
 $y = -\frac{x}{4} + \frac{1}{4}$

(b) $e^{2 \ln 4} = e^{\ln 16} = 16$ [1]

(c) $x = t^3 - 15t^2 + 48t - 25$

(i) $\frac{dx}{dt} = 3t^2 - 30t + 48$ [2]

$\frac{d^2x}{dt^2} = 6t - 30$ ✓

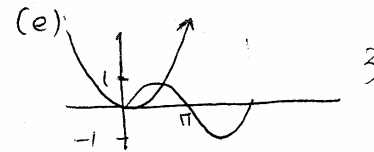
(ii) When $t=0, x=-25$ ✓

[3] $V=48, a=-30$ ✓

(iii) $v=0$ when $\frac{dx}{dt}=0$

$3(t^2 - 10t + 16) = 0$
 $(t-8)(t-2) = 0$
 $\Rightarrow t=2, 8$ [2]

(d) $\int_0^{\pi/8} \sec 2x \tan 2x dx$
 $= \frac{1}{2} [\sec 2x]_0^{\pi/8}$ [2]
 $= \frac{1}{2} (\sec \frac{\pi}{4} - \sec 0) = \frac{\sqrt{2}-1}{2}$



Area is bounded
 by $y = 1.5 - x^2$

SECTION B

Question 3

(a) (i) $y = e^{-x} \tan x$

$$\begin{aligned} \frac{dy}{dx} &= e^{-x} \sec^2 x + \tan x (-e^{-x}) \\ &= e^{-x} (\sec^2 x - \tan x) \quad [2] \end{aligned}$$

(ii) $y = \ln(x^4) + (\ln x)^4$
 $= 4 \ln x + (\ln x)^4$

$$\begin{aligned} \frac{dy}{dx} &= \frac{4}{x} + 4 \cdot (\ln x)^3 \cdot \frac{1}{x} \\ &= \frac{4}{x} (1 + (\ln x)^3) \quad [2] \end{aligned}$$

(iii) $y = \frac{x^2}{\ln x}$

$$\begin{aligned} \frac{dy}{dx} &= \frac{\ln x \cdot 2x - x^2 \cdot \frac{1}{x}}{(\ln x)^2} \\ &= \frac{2x \ln x - x}{(\ln x)^2} \quad [2] \end{aligned}$$

(b) (i) $y = e^{x/2}$

$$y' = \frac{1}{2} e^{x/2}$$

$$m(3) = \frac{1}{2} e^{-3/2}$$

∴ Tangent is

$$y - y_1 = m(x - x_1)$$

$$1 - e^{3/2} = \frac{1}{2} e^{3/2} (x - 3)$$

$$2y - 2e^{3/2} = x e^{3/2} - 3e^{3/2}$$

$$x e^{3/2} - 2y - e^x = 0$$

$$\text{or } y = \frac{1}{2} e^{3/2} (x - 1) \quad [2]$$

(ii) When $x = 0$, $y = 0$

$$\therefore A \text{ is } (1, 0) \quad [1]$$

$$\begin{aligned} \text{(iii) Area} &= \int_0^3 e^{x/2} dx - \Delta ABP \\ &= [2e^{x/2}]_0^3 - \frac{1}{2} \times 2 \times e^{3/2} \\ &= 2e^{3/2} - 2 - e^{3/2} \\ &= e^{3/2} - 2 \text{ units}^2 \end{aligned}$$

[2]

(c) $\int \left(\frac{e}{x} + \frac{x}{e} \right) dx$

$$= e \ln x + \frac{x^2}{2e} + C \quad [2]$$

(d) $\frac{d}{dx} \ln(e^{2x})$

$$= \frac{d}{dx} [\ln e^2 + 2 \ln x]$$

$$= \frac{2}{x}$$

[2]

Question 4

(a) $H = 4 + 2 \cos \frac{\pi t}{6}$, $t > 0$

(i) $\frac{dH}{dt} = 0 + (2 \sin \frac{\pi t}{6}) \times \frac{\pi}{6}$
 $= -\frac{\pi}{3} \sin \frac{\pi t}{6}$

When $t = 10$

$\frac{dH}{dt} = -\frac{\pi}{3} \sin \frac{10\pi}{6}$
 $= -\frac{\pi}{3} \left(-\frac{\sqrt{3}}{2}\right)$

$= \frac{\sqrt{3}\pi}{6} = \frac{\pi}{2\sqrt{3}}$

$\therefore$ Rate of change is $\frac{\pi}{2\sqrt{3}}$ m/hr
 $[\approx 0.9069]$ [2]

(ii) Increasing, because rate is positive, indicating tide rising. [1]

(b) (i) $\frac{dP}{dt} > 0$ [1] (ii) $\frac{d^2P}{dt^2} < 0$ [1]

(c) (i) Given $\frac{dM}{dt} = -kM$ $k > 0$

Consider $M = M_0 e^{-kt}$

$\frac{dM}{dt} = M_0 (-k e^{-kt})$
 $= -k(M_0 e^{-kt})$
 $= -kM$ as required [1]

(ii) $\frac{1}{2} M_0 = M_0 e^{-k \times 28}$

$\therefore \ln \frac{1}{2} = -28k$

$\therefore k = \frac{\ln \frac{1}{2}}{-28}$

$\approx 0.024755 \dots$

≈ 0.02476 [2]

(iii) $\frac{M_0}{100} = M_0 e^{-kt}$

$\therefore \ln \frac{1}{100} = -kt$

$t = \frac{\ln \frac{1}{100}}{-k}$

$= \frac{\ln 100}{k}$

≈ 186.03 [2]

$\approx 186 \text{ yrs } 10 \text{ days}$

(d) $f(x) = x^{-e} e^x$, $x > 0$

(i) $f'(x) = x^{-e} \cdot e^x + e^x \cdot (-e x^{-e-1})$
 $= e^x (x^{-e} - e x^{-e-1})$
 $= e^x x^{-e-1} (x - e)$ [3]

(ii) $f'(e^-) = (+)(+)(-)$
 < 0

$f'(e^+) = (+)(+)(+)$
 > 0

$\therefore$ Turning Point is a Relative Minimum.

[2]

SECTION C

Question 5

(a) $v = e^{3t} - 4e^t$

(i) initial velocity = $e^{3(0)} - 4e^{(0)}$
= $1 - 4$
= -3 m/s
= 3 m/s (1) in the left direction

$$a = \frac{dv}{dt}$$

$$= 3e^{3t} - 4e^t$$

initial acceleration = $3e^{3(0)} - 4e^{(0)}$
= $3 - 4$

$$= -1 \text{ m/s}^2$$
$$= 1 \text{ m/s}^2 \text{ to the left. (1)}$$

(ii) particle is stationary when $v=0$.

$$0 = e^{3t} - 4e^t$$

$$e^{3t} = 4e^t$$

$$e^{2t} = 4$$

(take lns of both sides)

$$2t = \ln 4$$

$$t = \frac{1}{2} \ln 4$$

$$t = \ln 4^{\frac{1}{2}}$$

$$\underline{t = \ln 2 \text{ seconds.}}$$

(2)

$$(iii) \quad v = \frac{dx}{dt} = e^{3t} - 4e^t$$

$$x = \int (e^{3t} - 4e^t) dt$$

$$x = \frac{1}{3} e^{3t} - 4e^t + C \quad \text{when } t=0 \quad x=0$$

$$0 = \frac{1}{3} e^{3(0)} - 4e^{(0)} + C$$

$$0 = \frac{1}{3} - 4 + C$$

$$C = \frac{11}{3}$$

(2)

$$\therefore x = \frac{1}{3} e^{3t} - 4e^t + \frac{11}{3}$$

$$(iv) \quad \text{when } t = \ln 4$$

$$x = \frac{1}{3} e^{3(\ln 4)} - 4e^{(\ln 4)} + \frac{11}{3}$$

$$x = \frac{1}{3} e^{\ln 4^3} - 4e^{\ln 4} + \frac{11}{3}$$

$$x = \frac{1}{3} \cdot 64 - 4 \cdot 4 + \frac{11}{3}$$

$$x = 9 \text{ metres to the right of the origin}$$

(1)

$$(v) \quad \text{Find displacement after } t = \ln 2 \text{ seconds}$$

$$x = \frac{1}{3} e^{3(\ln 2)} - 4e^{(\ln 2)} + \frac{11}{3}$$

$$x = \frac{1}{3} e^{\ln 2^3} - 4e^{\ln 2} + \frac{11}{3}$$

$$x = \frac{1}{3} \cdot 8 - 4 \cdot 2 + \frac{11}{3}$$

$$x = -\frac{5}{3} \text{ m}$$

$$x = \frac{5}{3} \text{ m to the left of the origin.}$$

$\therefore$ distance travelled in the first $\ln 4$ seconds

$$= \frac{5}{3} + \left(9 - \frac{5}{3}\right)$$

$$= \frac{37}{3} \text{ m.}$$

$$= 12\frac{1}{3} \text{ m.}$$

(2)

(b) $u(x) = R - 1 + e^x$
 $v(x) = R + 1 - e^x$

$$(i) A = \int_0^{\ln 2} [u(x) - v(x)] dx$$

$$A = \int_0^{\ln 2} (R - 1 + e^x - (R + 1 - e^x)) dx$$

$$A = \int_0^{\ln 2} (-2 + 2e^x) dx$$

$$A = 2 \int_0^{\ln 2} (e^x - 1) dx$$

$$A = 2 [e^x - x]_0^{\ln 2}$$

$$A = 2 [(e^{\ln 2} - \ln 2) - (e^0 - 0)]$$

$$A = 2 [2 - \ln 2 - 1]$$

$$A = 2 (1 - \log_e 2)$$

(3)

$$\begin{aligned}
 \text{(ii)} \quad V &= \pi \int_0^{\ln 2} \left((R-1+e^x)^2 - (R+1-e^x)^2 \right) dx \\
 &= \pi \int_0^{\ln 2} \left((R-1+e^x)(R-1+e^x) - (R+1-e^x)(R+1-e^x) \right) dx \\
 &= \pi \int_0^{\ln 2} \left(R^2 - R + Re^x - R + 1 - e^x + Re^x - e^x + e^{2x} \right. \\
 &\quad \left. - (R^2 + R - Re^x + R + 1 - e^x - Re^x - e^x + e^{2x}) \right) dx \\
 &= \pi \int_0^{\ln 2} (-4R + 4Re^x) dx \\
 &= 4\pi R \int_0^{\ln 2} (e^x - 1) dx \\
 &= 2\pi R \cdot 2 \int_0^{\ln 2} (e^x - 1) dx \\
 &= 2\pi R A
 \end{aligned}$$

from (i) $A = 2 \int_0^{\ln 2} (e^x - 1) dx$

3

Section C

Question 6

$$\begin{aligned}
 \text{(a)} \quad & \int_{-1}^1 \left(2x - \frac{1}{x-2} \right) dx \\
 &= \left[x^2 - \ln|x-2| \right]_{-1}^1 \\
 &= (1^2 - \ln|-1|) - ((-1)^2 - \ln|-1-2|) \\
 &= 1 - 0 - (1 - \ln 3) \\
 &= \ln 3
 \end{aligned}$$

NB $\ln x$ is **ONLY** defined for $x > 0$

ALTERNATIVELY using point symmetry about $x = 2$. Draw a diagram!

$$\begin{aligned}
 \int_{-1}^1 \left(2x - \frac{1}{x-2} \right) dx &= \int_{-1}^1 2x \, dx - \int_{-1}^1 \frac{dx}{x-2} \\
 &= \left[x^2 \right]_{-1}^1 + \int_3^5 \frac{dx}{x-2} \\
 &= 0 + [\ln(x-2)]_3^5 \\
 &= \ln 3 - \ln 1 \\
 &= \ln 3
 \end{aligned}$$

NB

$$-\int_{-1}^1 \frac{dx}{x-2} = +\int_3^5 \frac{dx}{x-2}$$

$$(b)(i) f(x) = e^{3x} \sin 2x$$

$$f'(x) = e^{3x} \cdot 2 \cos 2x + 3e^{3x} \sin 2x$$

$$f'(x) = 2e^{3x} \cos 2x + 3e^{3x} \sin 2x$$

$$\text{let } f'(x) = 0$$

$$2e^{3x} \cos 2x + 3e^{3x} \sin 2x = 0$$

$$e^{3x} (2 \cos 2x + 3 \sin 2x) = 0$$

$$e^{3x} \neq 0, \quad 2 \cos 2x + 3 \sin 2x = 0$$

$$3 \sin 2x = -2 \cos 2x$$

$$\frac{3 \sin 2x}{\cos 2x} = -2$$

$$3 \tan 2x = -2$$

$$\tan 2x = -\frac{2}{3}$$

$$\therefore f'(x) = 0 \text{ when } \tan 2x = -\frac{2}{3}$$

(2)

$$(ii)(i) \text{ let } f(x) = 0$$

$$e^{3x} \sin 2x = 0$$

$$e^{3x} \neq 0 \quad \sin 2x = 0$$

$$2x = 0, \pi, 2\pi, \dots$$

$$x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}, \dots$$

$$\therefore B = \frac{\pi}{2} \text{ and } C = \pi$$

(2)

$$(B) \text{ from (i) } f'(x) = 0 \text{ when } \tan 2x = -\frac{2}{3}$$

acute angle

$$\tan \alpha = \frac{2}{3}$$

$$\alpha = 0.588$$

$$2x = \pi - \alpha, 2\pi - \alpha, \dots$$

$$x = \frac{\pi - \alpha}{2}, \frac{2\pi - \alpha}{2}, \dots$$

$$x = 1.28$$

the x-coordinate of A, correct to 2 dec. places.

(2)

$$(iii) (a) g(x) = \frac{1}{13} (3e^{3x} \sin 2x - 2e^{3x} \cos 2x)$$

$$\begin{aligned} g'(x) &= \frac{1}{13} (3e^{3x} \cdot 2\cos 2x + 9e^{3x} \sin 2x - (2e^{3x} \cdot 2\sin 2x) + 6e^{3x} \cos 2x) \\ &= \frac{1}{13} (6e^{3x} \cos 2x + 9e^{3x} \sin 2x + 4e^{3x} \sin 2x - 6e^{3x} \cos 2x) \\ &= \frac{1}{13} (13e^{3x} \sin 2x) \\ &= e^{3x} \sin 2x \quad (3) \end{aligned}$$

$$(B) A = \int_0^{\frac{\pi}{2}} e^{3x} \sin 2x \, dx$$

$$= \frac{1}{13} [3e^{3x} \sin 2x - 2e^{3x} \cos 2x]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{13} [3e^{\frac{3\pi}{2}} \sin 2(\frac{\pi}{2}) - 2e^{\frac{3\pi}{2}} \cos 2(\frac{\pi}{2}) - (3e^{3(0)} \sin 2(0) - 2e^{3(0)} \cos 2(0))]$$

$$= \frac{1}{13} [3e^{\frac{3\pi}{2}} \sin \pi - 2e^{\frac{3\pi}{2}} \cos \pi - (3e^0 \sin 0 - 2e^0 \cos 0)]$$

$$= \frac{1}{13} [3e^{\frac{3\pi}{2}}(0) - 2e^{\frac{3\pi}{2}}(-1) - (3(1)(0) - 2(1)(1))]$$

$$= \frac{1}{13} (2e^{\frac{3\pi}{2}} + 2)$$

$$= \frac{2}{13} (e^{\frac{3\pi}{2}} + 1) \text{ units}^2 \quad (3)$$